# Traffic Sign Classification Under Varying Environmental Conditions

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***Abstract*—Reliable traffic sign recognition is essential for intelligent transportation systems, particularly when images are affected by challenging environmental conditions. This work investigates the performance of two machine learning classifiers on the German Traffic Sign Recognition Benchmark (GTSRB) dataset using visual representations extracted from a pretrained DINOv2 model. The extracted features are used to train a Multi-Layer Perceptron (MLP) and a Linear Support Vector Machine (SVM), with the dataset divided into training and validation subsets in an 80:20 ratio. To examine model robustness, the evaluation considers 14 categories of image corruption across multiple severity levels.While the Linear SVM records an accuracy of 90.67% and a macro F1-score of 0.9203, the MLP achieves an accuracy of 96.30% and a macro F1-score of 0.9656. The classification performance of both methods is compared, and the impact of various types and degrees of image contamination on traffic sign recognition is examined.**

***Index Terms*—Traffic sign classification, image corruption, robustness, DINOv2, multilayer perceptron, support vector machine**

## I. Introduction

In intelligent transportation systems and self-driving cars, traffic sign recognition is crucial. It aids cars in recognizing traffic signs and adhering to regulations. However, real-world factors like fog, snow, dim lighting, blur, and noise make it harder to identify traffic signs in photos.

This study uses the GTSRB dataset to investigate how various circumstances impact the classification of traffic signs. After extracting features from traffic sign photos using a pretrained DINOv2 model, MLP and Linear SVM models are utilized for classification. Clean photos and 14 different kinds of image corruption at five different severity levels are used to test the models. To assess their performance and determine how well they manage various degrees of corruption, accuracy and macro F1-score are utilized.

## II. Related Work

Both deep learning and conventional machine learning methods have been used to study traffic sign recognition. Models for classifying traffic signs are frequently assessed using the GTSRB dataset. More modern methods, like pretrained vision models, can minimize the need to train a model from scratch by extracting valuable image features.

A pretrained vision model called DINOv2 uses images to learn visual characteristics. For picture identification tasks, these attributes can be used to various classifiers. However, classification ability can be impacted by visual distortions such noise, blur, fog, and brightness variations. The evaluation of DINOv2-based features using MLP and Linear SVM classifiers under various corruption types and severity levels is the main goal of this study.

## III. Proposed System Architecture

The GTSRB dataset is loaded and split into training and validation sets. The images are preprocessed and run through DINOv2-Small to produce 384-dimensional feature vectors, which are then used to train two classifiers, MLP and Linear SVM, whose performance is first evaluated on clean validation images. The proposed system assesses traffic sign classification under various image conditions using features extracted from a pretrained vision model.

Fourteen different kinds of image corruption are used at five different severity levels to test robustness. The trained classifier is then used to assess the corrupted images, and accuracy and macro F1-score are used to record the outcomes. Lastly, confusion matrices and performance graphs are used to examine the results. The study may compare clean-image performance with performance under various kinds and degrees of picture deterioration thanks to this approach.

## IV. Dataset and Methodology

### A. Dataset

Traffic sign categorization under various visual circumstances is investigated using the German Traffic Sign Recognition Benchmark (GTSRB) dataset. It includes pictures from 43 different types of traffic signs. The photos are normalized and scaled to $224 \times 224$ pixels before to feature extraction. Using a stratified 80:20 split with random seed 42, the dataset is split into training and validation sets. In both sets, this partition keeps the distribution of traffic sign classes comparable. The 7,842 photos in the validation set are used to assess how well the trained classifiers perform.

### B. Feature Extraction

This study extracts visual information from photos of traffic signs using a pretrained DINOv2-Small model. The approach is used to transform each image into a 384-dimensional feature vector rather than training an image recognition network from scratch. These vectors identify visual patterns that aid in differentiating between different categories of traffic signs. There is less need to run the same images through the vision model again because the extracted features are stored and utilized during classifier training and testing.

### C. Classification Models

A Linear Support Vector Machine (SVM) and a Multi-Layer Perceptron (MLP) are two classification techniques that are taken into consideration. Both models receive the feature vectors created by DINOv2. The MLP uses fully linked layers with 256 and 128 hidden units, ReLU activation functions, and dropout to lessen dependence on individual neurons during training. Its output layer predicts one of the 43 traffic sign classes. The Linear SVM provides a second classification approach for comparison. Evaluating both models on the same extracted features helps determine how the choice of classifier affects recognition performance.

### D. Image Corruptions

Traffic sign images can lose important visual details because of changes in weather, lighting, image quality, and camera movement. To examine these effects, this study considers 14 corruption types, each with five severity levels. The selected corruptions are Gaussian noise, impulse noise, shot noise, snow, fog, frost, motion blur, zoom blur, defocus blur, brightness changes, contrast changes, pixelation, elastic transformation, and JPEG compression. Each corruption is applied to the validation images at severity levels from 1 to 5. The resulting predictions are compared with the clean-image baseline to measure how image degradation affects classification performance.

### E. Gaussian Blur and Image Sharpening

Three traffic sign images are used to illustrate the visual effects of Gaussian blur and basic image sharpening. Fig. 1 compares the original, blurred, and sharpened images.

## V. Experimental Setup

An 80:20 ratio is used to separate the GTSRB dataset into training and validation sets. 384-dimensional feature vectors are extracted from the traffic sign photos by a pretrained DINOv2-Small model. Two classifiers, a Multi-Layer Perceptron (MLP) and a Linear Support Vector Machine (SVM), are trained using these features.

To determine baseline performance, the classifiers are assessed using clean validation images. After then, 70 corrupted conditions are created by applying 14 different types of image corruption at five different severity levels in order to assess robustness. The macro F1-score and accuracy are used to compare the impact of various corruptions and assess classification performance.




Fig. 1. Original, Gaussian-blurred, and sharpened traffic sign images.

The project is constructed in Python, and PyTorch is utilized for MLP training and feature extraction. During evaluation, the trained models and extracted characteristics are stored for later use. Dataset preparation, feature extraction, model training, corruption creation, and performance analysis are all handled by different scripts. Graphs and confusion matrices are created to aid in the interpretation of the experimental results, and the evaluation results are saved in a CSV file.

## VI. Evaluation Metrics

Accuracy and macro F1-score are used to assess the classifiers' performance. While the macro F1-score assigns equal weight to every traffic sign class, accuracy quantifies the proportion of correctly categorized photos.

The formula for accuracy is:

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Predictions}} \times 100 \tag{1}$$

Precision and recall are used to calculate the F1-score:

$$\text{Precision} = \frac{TP}{TP + FP} \tag{2}$$

$$\text{Recall} = \frac{TP}{TP + FN} \tag{3}$$

The F1-score is calculated as:

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{4}$$

The macro F1-score is obtained by averaging the F1-scores of all classes:

$$\text{Macro F1} = \frac{1}{C} \sum_{i=1}^{C} F1_i \tag{5}$$

Here, $TP$ represents true positives, $FP$ represents false positives, $FN$ represents false negatives, and $C$ is the total number of classes.

## VII. Results and Discussion

Using both clean and corrupted traffic sign images, the effectiveness of the MLP and Linear SVM classifiers is assessed. The trials look at how well both models perform in classification and how various image corruptions impact recognition accuracy.

### A. Clean-Image Classification Performance

The effect of image corruptions is then analyzed using accuracy and macro F1-score results across different corruption types and severity levels; confusion matrices are also examined to understand the classification errors made by each model; both classifiers are first evaluated on clean validation images to establish a baseline for the corruption experiments. The MLP achieves a validation accuracy of 96.30% and a macro F1-score of 0.9656, while the Linear SVM achieves 90.67% accuracy and a macro F1-score of 0.9203.

### B. Training Performance and Generalization

An overview of the classifiers' performance on unseen photos and their ability to learn from the retrieved features is given by the training and validation data. 99.38% training accuracy and 96.30% validation accuracy are attained by the MLP. The macro F1-scores for training and validation are 0.9956 and 0.9656, respectively. With corresponding macro F1-scores of 0.9816 and 0.9203, the Linear SVM achieves training accuracy of 96.43% and validation accuracy of 90.67%.

Both classifiers perform better on the training data than on the validation data. The gap is larger for the MLP in absolute accuracy terms, however this alone does not show serious overfitting. The results imply that validation performance should be included with training performance when comparing the two classifiers.

### C. Comparison of Classification Models

The MLP model achieves 96.30% validation accuracy with a macro F1-score of 0.9656. The Linear SVM model achieves 90.67% validation accuracy with a macro F1-score of 0.9203. These results show that MLP performs better on the validation dataset.

### D. Effect of Image Corruptions

The evaluation shows that different types of image corruption affect classification performance differently. Defocus blur and pixelation have relatively high average accuracy, while frost, snow, fog, and noise cause greater performance drops. This indicates that the type of corruption plays an important role in traffic sign recognition.

### E. Comparison Across Corruption Types

The average results across five severity levels show that different corruptions have different effects on traffic sign classification. Defocus blur achieves the highest average accuracy at 91.07%, followed closely by pixelation at 91.00%. Motion blur and brightness changes also produce comparatively better results than several other corruption types.

Frost, on the other hand, yields the lowest average accuracy (26.69%). The average accuracy for snow and fog is 36.29% and 38.37%, respectively. Performance is additionally lowered by noise-based corruptions; Gaussian noise achieves an average accuracy of 44.55%, shot noise 48.20%, and impulse noise 38.70%.

These results suggest that certain visual distortions are more sensitive to the classifier than others. Classification may become more challenging if corruptions significantly alter the visibility or look of traffic signs. In addition to accuracy, the macro F1-score results include further details on performance across courses.

### F. Performance Across Severity Levels

By assessing the MLP classifier at five levels for each form of corruption, the impact of corruption severity is investigated. The findings demonstrate that classification accuracy can be lowered by more severe image deterioration. For instance, accuracy drops from 68.77% at severity level 1 to 17.21% at severity level 5 under Gaussian noise. This suggests that the classifier has a harder time accurately identifying traffic signs when there is a lot of noise.

Nevertheless, not all forms of corruption and degrees of severity experience the same decline in performance. Depending on how each corruption impacts an image's visual characteristics, some variations might happen. The variations in classification accuracy over severity levels are shown in Fig. 2, which aids in determining the circumstances in which the classifier loses reliability.

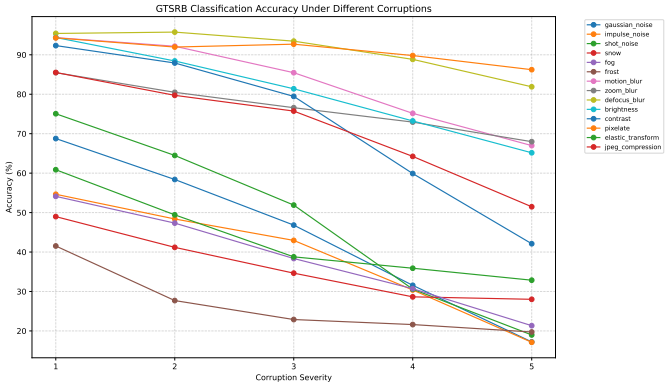



Fig. 2. MLP classification accuracy across corruption severity levels.

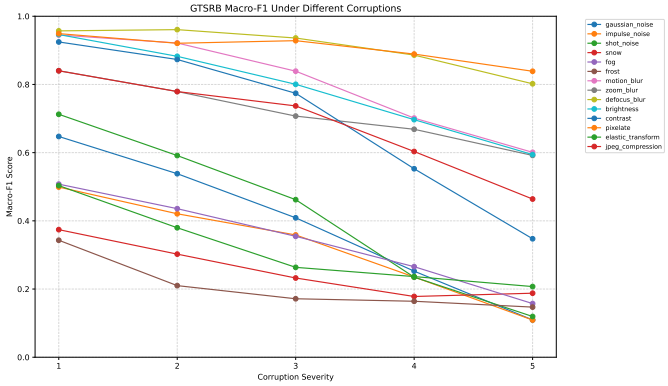



Fig. 3. MLP macro F1-score across corruption severity levels.

### G. Confusion Matrix Analysis

A class-wise view of the classification outcomes on clean validation images is given by the confusion matrices. They display the classes of traffic signs that are appropriately identified as well as the classes that are most commonly confused with one another. The confusion matrices for the Linear SVM and MLP classifiers are displayed in Figures 4 and 5, respectively.

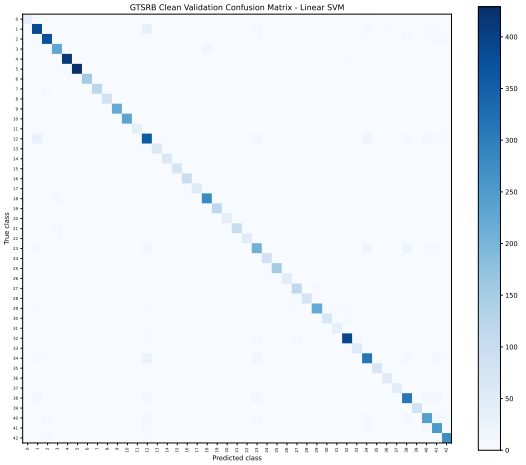



Fig. 4. Confusion matrix for clean-image classification using Linear SVM.

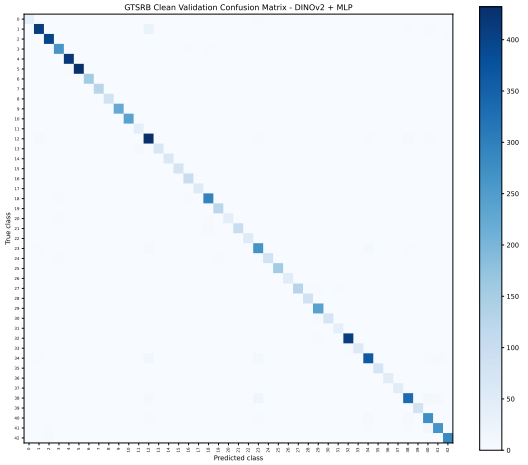



Fig. 5. Confusion matrix for clean-image classification using MLP.

## VIII. Conclusion

In this study, pretrained DINOv2 features with MLP and Linear SVM classifiers are used to assess traffic sign classification. On the GTSRB dataset, the MLP outperforms the Linear SVM in terms of validation accuracy. Additionally, the trials demonstrate that different image corruptions have distinct effects on categorization performance. Recognition is more difficult in situations like frost, snow, fog, and noise. All things considered, this study emphasizes how crucial it is to assess traffic sign classification models in various environmental settings.

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